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Understanding UK public preferences and attitudes to a decarbonized heating system

Substantial heterogeneity is evident in stances towards the development of aquifer thermal energy storage (ATES) systems in the UK. An assessment of individual preferences reveals substantial valuations of specific ATES characteristics that are nevertheless modified by latent attitudes regarding systemic and local benefits, safety concerns, information provision, and policy support.

This is a summary of:
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The problem

Heating of buildings consumes considerable energy and accounts for up to 20% of the UK's greenhouse gas emissions[1]. Decarbonization of heating systems is, therefore, essential if the country is to achieve its ambitious climate policy objectives. Aquifer thermal energy storage (ATES) systems use underground aquifers to store thermal energy and can support the transition to low-carbon systems by providing a sustainable source of heating and cooling while simultaneously reducing reliance on fossil fuels. Although extensively used in other parts of the world[2], ATES remains underused in the UK[3]. Enabling a responsible and equitable transformation of heating systems through the adoption of ATES while avoiding the opposition encountered by some renewable energy innovations[4] requires understanding the attitudes and concerns of the UK public over this technology, as well as insight into the preferences and priorities of local residents relating to the technical characteristics of ATES deployments.

The observation

We focused on the Greater Manchester area in England, a region with favourable geology for ATES deployment. We conducted an online survey of 1,758 individuals selected (by quota sampling) to be representative of the UK population that collected information on concerns and perceptions regarding ATES deployment in local public-use buildings, such as schools, hospitals and offices. The survey included a discrete choice experiment, which elicited choices across various specifications of ATES installations to evaluate respondents' preferences and the value they assign to the speed of ATES development, the opportunity for private households to connect to a public-use ATES system, the potential of ATES installations to reduce CO_2 emissions, and proximity to the installations (to capture the disutility associated with disruption caused by nearby ATES development). We performed latent class analyses and estimated mixed multinomial logit and hybrid choice models.

Our results point to considerable heterogeneity in stances towards ATES among survey respondents. We find evidence of four distinct attitudinal classes, ranging from cautiously negative to enthusiastically supportive (Fig. 1). Around 40% of our UK sample are sceptical towards ATES development and a further 40% are indifferent or undecided (weakly supportive but lacking conviction). At the other end of the spectrum, about 20% are enthusiastic supporters of the technology who welcome government incentives and community involvement. Estimates from the choice models suggest that respondents value rapid ATES deployments, large CO_2 emissions reductions and access of private households to public-use ATES systems. Nevertheless, willingness to pay for these characteristics varies substantially across our sample and depends on latent attitudes about the local and systemic benefits of ATES, policy and financial support, safety and environmental concerns, and community engagement and education. We find that increased support for information initiatives is inversely associated with rapid deployment of the technology, and that concerns over its safety are associated with reduced willingness to pay for CO_2 emissions reduction.

The implications

Decarbonization of energy systems presents considerable societal and technological challenges. ATES systems must be designed and introduced in a responsible manner that addresses the concerns of local populations and is aligned with their preferences, if these systems are to be accepted and widely adopted. It is also important to anticipate the considerable variation that could exist in individuals' perceptions and preferences. Familiarizing the population with the features of the new technology and presenting ATES as part of a strategic and systematic suite of climate policies are both essential to mitigate against misconceptions and encourage their buy-in. Finally, designs that prioritize the capacity for private use, have positive environmental impacts, and are amenable to speedy deployment are likely to enjoy increased support.

Despite our efforts to mitigate the limitations associated with stated preference studies, our results might be influenced by hypothetical bias. Moreover, our results capture attitudes and preferences prevailing at a specific point in time and are liable to change as awareness of the technology increases.

Our work presents a first step towards understanding the preferences of UK populations in relation to the use of underground energy storage to decarbonize heating and cooling. We hope to continue this line of research by studying preferences relating to the use of other geothermal technologies and the demand for district heating systems.

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FROM THE EDITOR

"Several aspects of this work stood out to us, most notably the focus on ATES, which is a lesser studied area of research, but also the combined approach of a discrete choice experiment with a social survey to better understand public perceptions, attitudes and support for ATES." **Silvana Lakeman, Senior Editor, *Nature Energy*.**

FIGURE

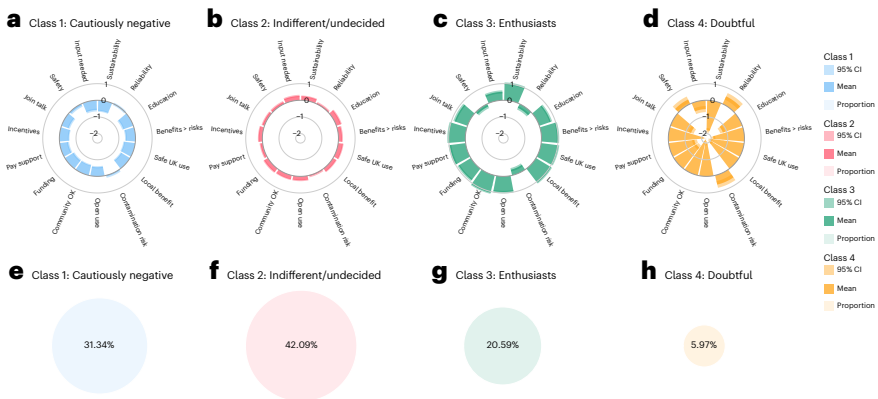

Fig. 1 | Four distinct stances towards ATES deployment identified among UK survey respondents. Charts (panels **a**–**d**) show predicted means and 95% confidence intervals for responses to a series of attitudinal statements, which are grouped into four statement classes. Each statement class represents a distinct stance towards ATES deployment. Circles (panels **e**–**h**) depict the predicted sample proportions for each stance. © 2026, Liu, T. et al., CC BY 4.0.

BEHIND THE PAPER

This work drew on research we conducted for the interdisciplinary collaborative project 'Aquifer thermal energy storage for decarbonisation of heating and cooling: overcoming technical, economic and societal barriers to UK deployment' (ATESHAC). The project aimed to study, characterize and ultimately develop strategies for overcoming technological and social barriers to the adoption of large-scale ATES deployments in the UK and other immature markets. From the early stages of project design, the team recognized the social dimensions of ATES development and placed responsible innovation at the core of the ATESHAC project. Our approach to achieving responsible and equitable ATES uptake combined the engagement of stakeholders with understanding of public awareness, perceptions, concerns, and even potential fears, that surround the introduction of novel heating and cooling technologies. We hope that this study will contribute to the uptake of low-carbon heating and cooling in the UK and beyond. **Y.K.**

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